A market such as Turkey or Vietnam, weakly correlated with the others, mechanically appears as a receiver. Trading hours add a second source of bias, since all funds trade in New York after the Asian markets have closed. The idiosyncratic network gives a different ranking (Spearman 0.30 with Diebold–Yilmaz): Taiwan and Indonesia are the main transmitters and Argentina is close to neutral. Argentina is thus exposed to the common emerging-market shock rather than to shocks from specific neighbors, and its largest country-specific shock, the August 2019 primary elections, does not turn it into a relevant transmitter in any of the four systems: in our data, that local shock did not spread.

Limitations. First, we study returns, not the realized volatilities of the base paper, so our networks describe the transmission of return shocks, not of risk. Second, ETF prices embed exchange-rate movements against the dollar and US trading hours, which affect the estimated directions. Third, the rolling windows of 104 weekly observations are short for the sparse estimators, whose results depend on tuning choices: NETS is selected by BIC over a grid of 15 values, which makes the series jump from one window to the next, and FNETS requires a positive-definite adjustment of the idiosyncratic covariance, gives an empty network in the full sample, and shows one-window drops that coincide with changes in the number of retained coefficients. With $N = 18$, the cross-section is also small for the factor-number criterion and for the consistency of the dynamic principal components, which rely on large N. Fourth, almost no lagged links survive in the sparse methods, so the Granger networks carry little information. Fifth, consecutive windows share 100 of their 104 weeks, so the stability measures are high partly by construction; they are informative for comparing methods but not as absolute values. Finally, we report no confidence intervals, so small differences, such as 81.0% against 80.0% between Diebold–Yilmaz and the lasso VAR, should not be interpreted as significant.

7 Conclusion

Measured connectedness among emerging equity markets depends less on the estimation method than on whether the common factor is removed. At the weekly frequency connectedness is high, around 80% for the dense measures, and almost entirely contemporaneous; the lasso VAR and Diebold–Yilmaz agree closely, and NETS reproduces the same pattern at a lower level. A single common factor explains more than half of the variance of returns, and removing it lowers connectedness by about 40%, a share that is stable across windows, horizons and samples; the peak of 2020 and the decline afterwards reflect changes in the weight of this factor rather than in bilateral links. Most of the co-movement of emerging markets is therefore common exposure, not contagion. What cannot be claimed with the same confidence is the direction of spillovers between countries, which our data barely identify, and the structure of the idiosyncratic network, which is unstable in short windows. Natural extensions are to study volatility connectedness, as in the base paper, to use higher-frequency data or longer windows for the factor-adjusted estimators, and to add more markets, which would also improve the estimation of the common factor.

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9 Data and Code Availability

The data (weekly ETF returns) and the full code are available at `https://github.com/ezellabra/ml4e-connectedness`. The pipeline is written in Python; NETS and FNETS are estimated in R (packages `nets` 0.9.1 and `fnets` 0.1.6) through a command-line bridge. All parameters are in `config.yaml` and the whole analysis is reproduced with `make all` (or `python run_all.py`).

Table 3. Robustness: mean total connectedness (%) by method, share of DY connectedness lost once the common factor is removed (FNETS), and FNETS common-variance share.

Scenario	Windows	DY	Lasso VAR	NETS	FNETS (idio.)	Lost w/o factor (%)	Common var. (%)
Baseline (w=104, H=4)	179	81.0	80.0	61.5	47.2	41	58
H=2	179	80.9	80.0	61.5	47.2	41	58
H=8	179	81.0	80.1	61.5	47.2	41	58
w=78	185	81.2	79.6	57.8	50.5	38	58
w=130	172	81.1	80.5	63.4	46.2	43	58
Excluding SPY	179	80.0	78.9	59.9	45.9	42	58
Excluding windows touching 2020	141	79.7	78.6	58.8	47.2	41	55

Table 4. Static connectedness: one estimate on the whole period (full sample) and on the subsamples before and after 2020. These are single long-sample estimates, not averages over rolling windows, so they differ from Tables 2 and 3.

Sample	Method	Total conn. (%)	LR p.corr. dens. (%)
Full sample	DY	81.9	100.0
Full sample	Lasso VAR	82.1	100.0
Full sample	NETS	73.5	60.1
Full sample	FNETS (idio.)	34.2	0.0
2011-2019	DY	83.0	100.0
2011-2019	Lasso VAR	82.8	100.0
2011-2019	NETS	69.6	60.1
2011-2019	FNETS (idio.)	34.1	2.0
2021-2026	DY	73.5	100.0
2021-2026	Lasso VAR	73.7	100.0
2021-2026	NETS	56.0	52.3
2021-2026	FNETS (idio.)	39.6	13.7

A Robustness

Table 3 reruns every method under alternative choices. The levels of Diebold–Yilmaz and lasso connectedness hardly move: between 79.7% and 81.2% and between 78.6% and 80.5% across all scenarios. NETS is the most sensitive to the window length (57.8% with 78-week windows, 63.4% with 130-week windows), as shorter windows leave the penalized estimator less information and make it keep fewer links. FNETS idiosyncratic connectedness ranges from 45.9% to 50.5%. The horizon matters very little: Diebold–Yilmaz changes by about 0.1 points between $H = 2$ and $H = 8$, and the sparse methods not at all, because their estimated lag coefficients are almost all zero and their variance decompositions are essentially driven by the contemporaneous covariance. Excluding the S&P 500 lowers Diebold–Yilmaz connectedness by one point, and removing every window that contains a week of 2020 (141 windows left) lowers Diebold–Yilmaz, lasso and NETS connectedness by one to three points and leaves FNETS unchanged, without changing the ranking of the methods.

The main quantities are stable across scenarios. The share of Diebold–Yilmaz connectedness lost once the common factor is removed lies between 38% (78-week windows) and 43% (130-week windows), against 41% in the baseline; the variance share of the common factor is 55–58%. The ordering of the comparison criteria is also unchanged: the correlation of total connectedness with Diebold–Yilmaz is at least 0.96 for the lasso and between 0.84 and 0.90 for NETS, and in every scenario the dense estimators are the most stable from one window to the next, followed by NETS and then FNETS.

The subsample estimates in Table 4 show a lower level of connectedness after 2020 than before: Diebold–Yilmaz connectedness falls from 83.0% in 2011–2019 to 73.5% in 2021–2026, and NETS from 69.6% to 56.0%. Idiosyncratic FNETS connectedness rises slightly (from 34.1% to 39.6%), while the variance share of the common factor falls from 58% to 47%. The decline after 2020 is therefore consistent with a weaker common component rather than with fewer idiosyncratic links. Figure 6 shows that, outside the windows that contain 2020, the time pattern of each method is the same as in the baseline.